

MATH 20250 HW4 HINTS AND SOLUTIONS

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1. Q3.3.2

An easier way of doing this problem is to write B as the product of A and another matrix, and then applying multiplicativity of determinants (LADW Theorem 3.3.5).

2. Q3.3.5

Nilpotence means that $A^k = 0$ for a certain natural number k . Some students might misread the definition as $A^k = 0$ for every natural number k , which is clearly the wrong definition.

3. Q3.3.9

Consider the Elementary Row operation (the third class, so that determinants are maintained):

$$\frac{1}{2} \begin{vmatrix} 1 & x_1 & y_1 \\ 1 & x_2 & y_2 \\ 1 & x_3 & y_3 \end{vmatrix} = \frac{1}{2} \begin{vmatrix} 1 & x_1 & y_1 \\ 0 & x_2 - x_1 & y_2 - y_1 \\ 0 & x_3 - x_1 & y_3 - y_1 \end{vmatrix} = \frac{1}{2} \begin{vmatrix} x_2 - x_1 & y_2 - y_1 \\ x_3 - x_1 & y_3 - y_1 \end{vmatrix}$$

The last expression is, by the geometric meaning of determinants, the area of the triangle with vertices $(0, 0)$, $(x_2 - x_1, y_2 - y_1)$, $(x_3 - x_1, y_3 - y_1)$. This triangle is a translation of the triangle with vertices (x_1, y_1) , (x_2, y_2) , (x_3, y_3) . Thus the two triangles have the same area. Thus, the area of the triangle in question is equal to the above expression.

4. Q3.3.10

It is quite doubtful one can write directly

$$\det \begin{bmatrix} I & * \\ 0 & A \end{bmatrix} = IA - 0*,$$

as if one dealing with 2-by-2 matrix. That equation is certainly correct, but considering the definition of determinant, it is not all that obvious: it is neither obvious from the properties of determinants listed in Section 3.3. Nor is it obvious from the definition of determinants in Equation 4.2 in Section 3.4. An argument has to be made for that equation.

4.1. Hard Approach. It is possible to approach the problem by Cofactor expansion and induction on the size of A or I , but a more intuitive way of doing this is by the formula:

$$\sum_{\sigma \in \text{Perm}(m+n)} \text{sgn}(\sigma) a_{1,\sigma(1)} \cdots a_{m+n,\sigma(m+n)}.$$

Here we suppose I is m -by- m and A is n -by- n or vice versa. Suppose we are dealing with (1). The above summation involving permutation amounts to every time selecting an entry in each row (or column), such that in every column (or row) there is only one entry selected, and multiplying all selected entries together. We can ignore any selection that would give us zero, so we avoid selecting any zero entry.

Since we are dealing with (1), then the lower-left quadrant is all zero. Thus, when we are selecting the entry for $m+1, \dots, m+n$ rows, we consider only the $m+1, \dots, m+n$ columns. (This amounts to the calculation of $\det(A)$.) Consequently, when we are selecting the entry for rows $1, \dots, m$, we have to be selecting from the $1, \dots, m$ columns. Since the upper-left quadrant is the identity matrix, for the i -th row we have to select the i -th column. Thus, we would have

$$\begin{aligned} \det \begin{bmatrix} I & * \\ 0 & A \end{bmatrix} &= \sum_{\sigma \in \text{Perm}(m+n)} \text{sgn}(\sigma) a_{1,\sigma(1)} \cdots a_{m+n,\sigma(m+n)} \\ &= \sum_{\sigma \in \text{Perm}(m+1, m+n)} \text{sgn}(\sigma) a_{m+1,\sigma(m+1)} \cdots a_{m+n,\sigma(m+n)} \\ &= \det(A). \end{aligned}$$

$\text{Perm}(m+1, m+n)$ refers to permutation of $m+1, \dots, m+n$.

The above argument, slightly modified, can be used directly to prove Q3.3.11.

4.2. Easy Approach. By transforming A inside

$$\begin{bmatrix} I & * \\ 0 & A \end{bmatrix}$$

into upper-triangular form, we get a larger upper-triangular matrix. Suppose the upper-triangular form of A is called A' . Then the new, larger upper-triangular matrix is

$$\begin{bmatrix} I & * \\ 0 & A' \end{bmatrix}$$

Because transforming a matrix into upper-triangular form involves only adding multiples of a row to another row, the transformation does not change the determinant. Thus,

$$\det(A) = \det(A')$$

$$\det \begin{bmatrix} I & * \\ 0 & A \end{bmatrix} = \det \begin{bmatrix} I & * \\ 0 & A' \end{bmatrix}$$

The product of the diagonal entries the larger upper-triangular matrix is equal to the product of a series of 1, and the diagonal entries of A' . Thus, we have

$$\det \begin{bmatrix} I & * \\ 0 & A' \end{bmatrix} = \det(A').$$

Thus, we conclude

$$\det \begin{bmatrix} I & * \\ 0 & A \end{bmatrix} = \det(A)$$

4.3. Easiest Approach. The easiest approach, which I did not come up with myself, provided by a student.

$$\begin{aligned} \begin{bmatrix} I & * \\ 0 & A \end{bmatrix} &= \begin{bmatrix} I & 0 \\ 0 & A \end{bmatrix} \begin{bmatrix} I & * \\ 0 & I \end{bmatrix} \\ \det \begin{bmatrix} I & * \\ 0 & A \end{bmatrix} &= \det \begin{bmatrix} I & 0 \\ 0 & A \end{bmatrix} \det \begin{bmatrix} I & * \\ 0 & I \end{bmatrix} = \det \begin{bmatrix} I & 0 \\ 0 & A \end{bmatrix} = \det(A). \end{aligned}$$

5. Q3.4.1: ABOUT PERMUTATION

Here, we consider permutation as permutation matrix.

When we talk about permutation here, when we point to a certain permutation, we are always actually (fundamentally) talking about the permutation matrices. For example, when we talk about the permutation matrix that turns 12345 to 12543, we are talking not about the specific action switching between the elements 3 and 5, but about the matrix

$$\begin{bmatrix} 1 & & & & \\ & 1 & & & \\ & & & & 1 \\ & & & 1 & \\ & & 1 & & \end{bmatrix}$$

This matrix switch between whatever is in position (3) and position (5) of a column vector. These can be elements from $\{1, 2, 3, 4, 5\}$ or $\{a, b, c, d, e\}$ or whatever

objects imaginable. The key is that the permutation is always done with regards to the positions, regardless of the specific elements that are permuted.

For example, in producing (54123), we have the following switching procedure:

$$(12345) \rightarrow (52341) \rightarrow (52143) \rightarrow (54123).$$

The second step should be said as switching between 3 and 5, because these are the positions switched, not as switching between 1 and 3. Only understood in the correct way, we can produce the correct reverse permutation:

$$(12345) \rightarrow (14325) \rightarrow (14523) \rightarrow (34521).$$

If we thought that we are switching the elements, and switched element 1 and element 3 in the second step, we would then thus get the wrong answer:

$$(12345) \rightarrow (14325) \rightarrow (34125) \rightarrow (34125) \rightarrow (54123).$$

6. Q3.4.2

Here is one way to look at this problem: because there are only finitely many permutations, when you have infinitely many $P^1, P^2, \dots$, there must be repetitions, say $P^i = P^j$, where $i < j$. Then we can conclude $P^{j-i} = I$.

7. Q3.5.5

One possible cofactor expansion is

$$\det(A) = \sum_{i=1}^n (-1)^{n+1} a_{i,1} \det(A_{i,1}).$$

(This is the cofactor expansion along the first column. Some may choose to use cofactor expansion along the first row. Both ways are correct.)

Thus, we have $D_n = \det(A) = \det(A_{1,1}) - \det(A_{2,1}) = D_{n-1} + D_{n-2}$.

Some students mis-remembered the formula as $\det(A) = \sum_{i=1}^n a_{i,1} \det(A_{i,1})$. In this case their computation would yield $D_n = D_{n-1} - D_{n-2}$.

8. Q3.7.1

The answer is *TTFFTTFTTT*.

8.1. **(c).** (c) should be False.

First of all, faced with such kind of question, the student should at least try verifying the statement for some small matrices.

Interchanging two rows is equivalent to multiplying the permutation matrix of one switch. The determinant of the latter is -1. Thus, interchanging two rows/columns negates the determinant of the matrix.

8.2. **(i).** (i) should be true. It is already known that if a matrix A is invertible then $\det(A)$ is non-zero. The student might be confused about the reverse direction.

Notice that elementary row transformation alters the determinant only by multiplying a non-zero factor. Thus it preserves the zero-ness or non-zero-ness of the determinant.

If a matrix is non-invertible, then its row echelon form has at least one empty row, meaning that its determinant is 0. Thus, (contrapositive) if the determinant of a matrix is non-zero, then the matrix is invertible.

9. Q3.7.2

Determinants are not additive: it is not necessary that

$$\det(A + B) = \det(A) + \det(B).$$

At least two students try to apply the above non-existent equation. They obviously would yield wrong answers.

To solve this problem, one can apply Property 12 from Summary 3.6 of Section 3.3 of LADW.

10. SUPPLEMENTARY PROBLEM 1

A tentative solution for reference here: there are 145 matrices with determinant 0, and 120 matrices with determinant 1, 2, 3 or 4, in each case, respectively.

We already know that there are 480 invertible matrices in $M_{\mathbb{Z}/5\mathbb{Z}}^2$. Thus, there are 145 non-invertible matrices. Since a matrix is invertible if and only if its determinant is non-zero, there are 145 matrices $A \in M_{\mathbb{Z}/5\mathbb{Z}}^2$ such that $\det(A) = 0$.

Consider we have A such that $\det(A) = 1$. We can get matrices of determinant 2, 3, 4 by multiplying the first row with 2, 3, 4. Conversely, for a matrix A with $\det(A) = 2, 3, 4$, we can multiply the first row of A with $\frac{1}{2} = 3, \frac{1}{3} = 2, \frac{1}{4} = 4$, and get matrix with determinant 1. There are equal number of matrices with determinants 1, 2, 3, 4. Thus, for $\det(A) = 1, 2, 3, 4$, there are $480/4 = 120$ possible cases for A , in each case.

11. SUPPLEMENTARY PROBLEM 2

For this problem, we can apply Equation 3.4.2 in LADW:

$$\det(A) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) a_{1,\sigma(1)} \dots a_{n,\sigma(n)}.$$

Here, S_n is the set of all permutations of $(1, \dots, n)$ and $\operatorname{sgn}(\sigma)$ is the sign of the permutation. The above formula gives (iii) immediately.

12. SUPPLEMENTARY PROBLEM 3

The answer should be -1008 .

In the cofactor expansion of determinant, the sign of the summation is alternating (a few student might forget this):

$$\det(A) = \sum_{i=1}^n (-1)^{i+1} a_{i,1} \det(A_{i,1}).$$

Thus, the function in question is actually

$$f(x) = -1008x + C$$

where C is constant. Thus, $f'(x) = -1008$.

Also, if we are not taking the cofactor expansion along the first column or row, then there might be an additional factor (-1) in the coefficients. This is because the i, j -th cofactor is defined as $(-1)^{i+j} \det(A_{i,j})$. Thus, if we are factorizing along the j -th column, we would have

$$\det(A) = \sum_{i=1}^n (-1)^{i+j} a_{i,j} \det(A_{i,j}).$$

13. SUPPLEMENTARY PROBLEM 4

In general, the student should first try to approach this problem by substituting the 100 in the problem by an unknown n , and try to let n be some small values like $n = 2, 3, 4$. This way they can avoid mistakes like claiming that columns of $C - I$ are linearly dependent.

13.1. (iii).

13.1.1. *Eigenvector Approach.* One way of doing this is to calculate the eigenvalues and eigenvectors of the matrix.

One basis of eigenvectors is $[1, 1, \dots, 1]^T$ and $[-1, \dots, n-1, \dots, -1]^T$ where $n-1$ is the i -th entry, and $i = 1, \dots, n-1$. The respective eigenvalues are $n-1$ for the first and -1 for the rest of the vectors. Thus, the answer is $(-1)^{n-1}(n-1)$. When $n = 100$, the answer is -99 .

An easier way of explaining this is that by lining up the above column vectors, we get a matrix P . Then, $P^{-1}(C - I)P$ is a diagonal matrix, with diagonal entries $n-1, -1, \dots, -1$. Apply the fact that determinants are the same for similar matrices, and we get the answer.

13.1.2. *Elementary Row Action Approach.* One can also approach the problem by elementary row action. By subtracting every row with row 1, we get a matrix, whose first row and first column are all 1, diagonal entries are all -1 , and specially upper-left-most entry is 0.

$$\begin{bmatrix} 0 & 1 & \cdots & 1 \\ 1 & -1 & & \\ \vdots & & \ddots & \\ 1 & & & -1 \end{bmatrix}$$

Adding the second to the n -th row to the first row we get:

$$\begin{bmatrix} n-1 & 0 & \cdots & 0 \\ 1 & -1 & & \\ \vdots & & \ddots & \\ 1 & & & -1 \end{bmatrix}$$

This is a lower triangular matrix, so its determinant is $(-1)^{n-1}(n-1)$.

13.2. (iv). For $C - I + D$, by subtracting every other row with the first row, we get a matrix whose first row entries are all 1, diagonal entries are $1, 1, 2, \dots, n-1$, and all other entries are 0. The determinant of such a matrix is $(n-1)!$. Thus, $\det(C - I + D) = (n-1)!$.